The conjectured generating function for OEIS A250757, proved

Adrian Perez Fontelles
Independent researcher

6 September 2026

Abstract

OEIS A250757 counts arrays of a fixed width under a local condition, and carries a conjectured generating function — and nothing else. It is true. The count is a walk count in a finite digraph on $S = 27$ vertices, so its generating function is a rational function whose numerator and denominator have degrees bounded by S; the conjectured expression is another rational function of known degree; and two rational functions of bounded degree agree as soon as enough coefficients of their expansions do. Comparing 35 coefficients in exact arithmetic therefore settles the conjecture, rather than testing it.

2020 Mathematics Subject Classification. 05A15, 05A19, 68Q45.

1 The conjecture

OEIS A250757 is "Number of (2+1) X (n+1) 0..2 arrays with nondecreasing x(i,j)-x(i,j-1) in the i direction and nondecreasing x(i,j)+x(i-1,j) in the j direction.". It has offset 1 and begins

$$105, 237, 423, 663, 957, 1305, \ \ldots$$

The entry states, as a conjecture contributed by a reader and never marked settled:

G.f.: 3*x*(35 - 26*x + 9*x^2) / (1 - x)^3.

As of the "Last modified" line on the live entry (Nov 18 2018 07:38:46, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved. The entry carries no conjectured recurrence, which is why this generating function is the whole of what is open here.

2 The count is a walk count

Every condition the entry names is decided inside a bounded number of consecutive rows: it compares a cell with cells a bounded distance from it, and no comparison reaches further than that. So a strip of that many consecutive rows is a state of a finite automaton, an array is read off row by row, and appending one more row is one step, legal exactly when the row it completes satisfies every condition that becomes testable.

Taking those strips as the vertices of a digraph G with adjacency matrix M, and writing ι for the indicator of the states an array may begin in and τ for those it may end in,

$$a(n) \;=\; \iota^{\top} M^{\,n-1+1} \tau,$$

and there are $S = 27$ states. The digraph is built by reading the entry's own English, and the model reproduces all 39 terms the entry publishes, exactly, in integer arithmetic — a misreading of the condition gives a different digraph and different counts, so this ties the model to the entry and not merely to the arithmetic.

The family this entry belongs to is read by one model, described by its author as: "'... arrays with nondecreasing <f>(x(i,j),x(i,j-1)) in the i direction and nondecreasing <g>(x(i,j),x(i-1,j)) in the j direction' and '... with nondecreasing <statistic> of every <k> consecutive values in every row and column'."

3 Its generating function is rational, with bounded degree

Lemma 1. *$A(x) = \sum_{n\geq 1} a(n)x^n = N_1(x)/D_1(x)$ where $D_1(x) = \det(I - xM)$ has degree at most S and $D_1(0) = 1$, and $\deg N_1 < S + 1$.*

Proof. Summing the geometric series of matrices, $\sum_{j\geq 0} M^j x^j = (I - xM)^{-1}$ as a formal power series, so $A(x)$ is $x^c\, \iota^\top (I - xM)^{-1}\tau$ for the constant c relating the walk index to the entry's own n. By Cramer's rule every entry of $(I - xM)^{-1}$ is a polynomial of degree less than S divided by $\det(I - xM)$, and $\det(I - xM)$ is a polynomial of degree at most S with constant term $\det I = 1$. □

Theorem 2. *The conjectured generating function is $A(x)$.*

Proof. Write the conjecture as N_2/D_2 with $\deg N_2 = 3$ and $\deg D_2 = 3$; its expansion as a power series exists because $D_2(0) \neq 0$. Put

$$R(x) \;=\; N_1(x)D_2(x) - N_2(x)D_1(x),$$

a polynomial of degree at most 30 by Lemma 1. The difference $A - N_2/D_2$ equals $R/(D_1D_2)$, and $D_1(0)D_2(0) \neq 0$, so if the two power series agree in every coefficient up to x^{30} then R has a zero of order greater than its own degree at the origin, which forces $R \equiv 0$ and hence $A = N_2/D_2$ identically.

The coefficients of A were produced from the model by repeated matrix–vector products in exact integer arithmetic, those of N_2/D_2 by exact rational long division, and 35 of them — more than the $30 + 1$ the argument requires — agree. □

4 A recurrence the entry does not state

The denominator of a rational generating function is a linear recurrence written backwards, so the theorem above yields one for free. With $D_2(x) = 1 - \sum_{i\geq 1} c_i x^i$ after normalising $D_2(0) = 1$, comparing coefficients of x^n in $A(x)D_2(x) = N_2(x)$ gives

$$a(n) \;=\; \sum_{i\geq 1} c_i\, a(n-i) \qquad \text{for every } n > 3,$$

a constant-coefficient linear recurrence of order 3, valid past the degree of the numerator. Its coefficients are the integers $c_1, \ldots, c_3$ read off D_2, the first few being $3,\ -3,\ 1$. The entry records no recurrence, so this is not a restatement of anything already there.

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces every one of the 39 published terms, with the index shift read off the entry's offset rather than fitted.

Second, the arrays themselves were enumerated for the smallest shapes the entry counts, directly from its English and with no automaton, and the counts agree. Writing that check is where the risk actually sits: an earlier version of it dropped a clause from the entry's name in three cases and disagreed with the model, and in all three the check was wrong and the model was right.

Third, the coefficient comparison was carried out over the whole range required by the degree bound rather than on a prefix, in exact arithmetic throughout.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A250757.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.